
9. The Directed Edge Task:

Objective: Determine whether two nodes are connected by a directed edge in either direction, using basic edge facts.

- Variable set: $X = \{X_1, X_2\}$
- Head variable set: $X^h = \{X_1, X_2\}$
- Body predicates:

$$b = \{\text{edge}(X_1, X_2), \text{d-edge}(X_2, X_1)\}$$

- Head predicate: $h = \text{d-edge}(X_1, X_2)$
- Background knowledge:

$$\mathbb{B} = \{\text{edge}(a, b), \text{edge}(b, d), \text{edge}(c, c), \dots\}$$

- Positive examples:

$$\mathcal{P} = \{\text{d-edge}(a, b), \text{d-edge}(b, a), \text{d-edge}(b, d), \text{d-edge}(d, b), \text{d-edge}(c, c), \dots\}$$

After propositionalization, ANDRE learns the following rules:

1. $\text{d-edge}(X_1, X_2) :- \text{d-edge}(X_2, X_1).$
2. $\text{d-edge}(X_1, X_2) :- \text{edge}(X_1, X_2).$

10. The Connectedness Task:

Objective: Learn the `connectedness` relation, which holds true if there exists a direct or transitive path (via one or more `edge` relations) between two nodes.

- Variable set: $X = \{X_1, X_2, X_3\}$
- Head variable set: $X^h = \{X_1, X_3\}$
- Auxiliary variable set: $X^a = \{X_2\}$
- Body predicates:

$$b = \{\text{edge}(X_1, X_3), \text{edge}(X_3, X_1), \text{edge}(X_2, X_3), \\ \text{connectedness}(X_1, X_2), \text{connectedness}(X_2, X_3)\}$$

- Head predicate: $h = \text{connectedness}(X_1, X_3)$
- Background knowledge:

$$\mathbb{B} = \{\text{edge}(a, b), \text{edge}(b, c), \text{edge}(c, d), \text{edge}(b, a), \dots\}$$

- Positive examples:

$$\mathcal{P} = \{\text{connectedness}(a, b), \text{connectedness}(b, c), \text{connectedness}(c, d), \\ \text{connectedness}(b, a), \text{connectedness}(a, c), \text{connectedness}(a, d), \\ \text{connectedness}(a, a), \text{connectedness}(b, d), \text{connectedness}(b, b), \dots\}$$

After applying the propositionalization strategy, the dataset E is generated and used to train ANDRE. The learned symbolic rules are as follows:

1. $\text{connectedness}(X_1, X_3) :- \text{edge}(X_1, X_3).$
2. $\text{connectedness}(X_1, X_3) :- \text{edge}(X_3, X_1).$
3. $\text{connectedness}(X_1, X_3) :- \text{connectedness}(X_1, X_2), \text{edge}(X_2, X_3).$

Table 6: Dataset statistics of Countries, Nations, UMLS, Kinship, UW-CSE, and Alzheimers-amine.

Dataset	#Constant	#Relation	#Raw Facts	# Trainable Samples	# Test Samples
Countries-S1	252	2	1,110	1,812,378	95,388
Countries-S2	252	2	1,063	1,197,092	299,272
Nations	14	56	2,364	36,496	1,920
UMLS	135	49	5,598	715,609	37,663
UW-CSE	1,209	15	2,675	3,861,667	429,074
Alzheimers	147	32	980	46,337	2,438

I ANDRE ON KNOWLEDGE BASES

Table 6 shows the statistical details of the knowledge bases used to evaluate the performance of ANDRE. The rest of appendix reports some symbolic rules extracted by ANDRE on knowledge-base reasoning benchmarks. For each task, we present representative rules discovered across different restarts and rule configurations. All rules are expressed in first-order logic form and extracted using the predicate identification procedure described in Section E. Only training and validation accuracy is reported here for brevity.

I.1 COUNTRIES DATASET

The *Countries* task aims to learn the target predicate `locatedIn( $X_1, X_2$ )`, using background relations such as `neighborOf` and recursive `locatedIn` facts. Below, we present all representative rules extracted by ANDRE, formatted in a prompt-style box for clarity.

Listing 2: Extracted Rules for `locatedIn( $X_1, X_2$ )` (Nations Dataset)

```
[1] locatedIn(X1, X2) :- locatedIn(X3, X2) and neighborOf(X3, X1).
    Training Acc: 0.9927 | Eval Acc: 0.9930

[2] locatedIn(X1, X2) :- locatedIn(X1, X3) and locatedIn(X3, X2).
    Training Acc: 0.9926 | Eval Acc: 0.9930

[3] locatedIn(X1, X2) :- locatedIn(X3, X1) and neighborOf(X2, X3).
    Training Acc: 0.9925 | Eval Acc: 0.9928

[4] locatedIn(X1, X2) :- locatedIn(X3, X1) and neighborOf(X3, X2).
    Training Acc: 0.9925 | Eval Acc: 0.9928

[5] locatedIn(X1, X2) :- locatedIn(X1, X3) and neighborOf(X3, X2).
    Training Acc: 0.9925 | Eval Acc: 0.9928

[6] locatedIn(X1, X2) :- neighborOf(X1, X3) and neighborOf(X2, X3).
    Training Acc: 0.9925 | Eval Acc: 0.9928

[7] locatedIn(X1, X2) :- locatedIn(X2, X3) and neighborOf(X3, X1).
    Training Acc: 0.9925 | Eval Acc: 0.9928
```

I.2 NATIONS DATASET

The *Nations* task aims to learn semantic relations between geopolitical entities using background predicates describing economic, political, and geographic interactions. Below, we report representative rules extracted by ANDRE for different target predicates.

`BLOCKPOSITIONINDEX( $X_1, X_2$ )`

The target predicate `blockpositionindex( $X_1, X_2$ )` captures hierarchical relationships between entities in the Nations knowledge base. The following rules were extracted by ANDRE:

Listing 3: Extracted Rules for `blockpositionindex( $X_1, X_2$ )` (Nations Dataset)

```
[1] blockpositionindex(X1, X2) :- blockpositionindex(X2, X1).
    Training Acc: 0.9590 | Eval Acc: 0.9620
    Val Coverage: (N_b=524, N_r=488, N_r/N_b=0.9313)
    Train Coverage: (N_b=9668, N_r=8920, N_r/N_b=0.9226)

[2] blockpositionindex(X1, X2) :-not(timesincewar(X1, X2)) and blockpositionindex(X2, X1).
```

```

Training Acc: 0.9474 | Eval Acc: 0.9474
Val Coverage: (N_b=486, N_r=455, N_r/N_b=0.9362)
Train Coverage: (N_b=9019, N_r=8383, N_r/N_b=0.9295)

[3] blockpositionindex(X1, X2) :- commonbloc0(X1, X2).
Training Acc: 0.8625 | Eval Acc: 0.8568
Val Coverage: (N_b=290, N_r=270, N_r/N_b=0.9310)
Train Coverage: (N_b=5394, N_r=5022, N_r/N_b=0.9310)

[4] blockpositionindex(X1, X2) :- commonbloc0(X2, X1).
Training Acc: 0.8627 | Eval Acc: 0.8536
Val Coverage: (N_b=284, N_r=264, N_r/N_b=0.9296)
Train Coverage: (N_b=5400, N_r=5028, N_r/N_b=0.9311)

[5] blockpositionindex(X1, X2) :- not(reintergovorgs(X2, X1)) and embassy(X2, X1) and not(
    commonbloc2(X1, X2)) and not(reltreaties(X1, X2)) and not(reldiplomacy(X1, X2)) and
    conferences(X2, X1).
Training Acc: 0.7538 | Eval Acc: 0.7474
Val Coverage: (N_b=44, N_r=42, N_r/N_b=0.9545)
Train Coverage: (N_b=803, N_r=742, N_r/N_b=0.9240)

```

Note. We refer to Appendix G for the definitions of $(N_r, N_b, \frac{N_r}{N_b})$.

INTERGOVORGS3(X_1, X_2)

The target predicate `intergovorgs3( $X_1, X_2$ )` models intergovernmental organization relationships in the Nations knowledge base. Below, we present representative rules extracted by ANDRE after removing redundant variants with unnecessary or overlapping literals.

Listing 4: Extracted Rules for `intergovorgs( $X_1, X_2$ )` (Nations Dataset)

```

[1] intergovorgs3(X1, X2) :- embassy(X1, X2) and ngoorgs3(X1, X2) and not(ngoorgs3(X2, X1))
    and not(exports3(X1, X2)) and not(releconomicaid(X2, X1)) and not(releconomicaid(X3, X1))
    and not(duration(X2, X1)) and not(lostterritory(X1, X3)).
Training Acc: 0.7466 | Eval Acc: 0.7482
Val Coverage: (N_b=648, N_r=529, N_r/N_b=0.8164)
Train Coverage: (N_b=12647, N_r=10135, N_r/N_b=0.8014)

[2] intergovorgs3(X1, X2) :- ngoorgs3(X1, X2) and not(economicaid(X2, X1)) and ngo(X2, X1) and
    not(reingo(X2, X1)) and not(relexportbooks(X1, X3)) and not(violentactions(X2, X1)) and
    not(warning(X1, X3)).
Training Acc: 0.6836 | Eval Acc: 0.6819
Val Coverage: (N_b=430, N_r=370, N_r/N_b=0.8605)
Train Coverage: (N_b=8473, N_r=7145, N_r/N_b=0.8433)

[3] intergovorgs3(X1, X2) :- not(accusation(X1, X2)) and intergovorgs(X1, X2) and ngo(X2, X1)
    and not(aidenemy(X2, X3)) and not(releconomicaid(X2, X1)) and not(expeldiplomats(X2, X1))
    and treaties(X2, X1) and not(lostterritory(X1, X3)).
Training Acc: 0.6867 | Eval Acc: 0.6806
Val Coverage: (N_b=466, N_r=387, N_r/N_b=0.8305)
Train Coverage: (N_b=8934, N_r=7420, N_r/N_b=0.8305)

[4] intergovorgs3(X1, X2) :- not(accusation(X2, X1)) and not(commonbloc2(X1, X2)) and reingo(
    X1, X2) and timesinceally(X1, X2) and not(relemigrants(X2, X3)) and not(lostterritory(X2,
    X3)).
Training Acc: 0.6551 | Eval Acc: 0.6733
Val Coverage: (N_b=347, N_r=322, N_r/N_b=0.9280)
Train Coverage: (N_b=6473, N_r=5737, N_r/N_b=0.8863)

[5] intergovorgs3(X1, X2) :- not(militaryalliance(X1, X2)) and intergovorgs(X1, X2) and not(
    expeldiplomats(X2, X3)) and not(reingo(X2, X1)) and not(relexportbooks(X1, X3)).
Training Acc: 0.6724 | Eval Acc: 0.6713
Val Coverage: (N_b=312, N_r=303, N_r/N_b=0.9712)
Train Coverage: (N_b=5949, N_r=5723, N_r/N_b=0.9620)

[6] intergovorgs3(X1, X2) :- not(commonbloc2(X1, X2)) and ngoorgs3(X1, X2) and reingo(X1, X2)
    and not(reingo(X2, X1)) and not(students(X1, X3)) and not(tourism(X2, X1)) and not(
    dependent(X1, X3)) and not(violentactions(X1, X2)) and not(severdiplomatic(X1, X3)).
Training Acc: 0.6517 | Eval Acc: 0.6468
Val Coverage: (N_b=349, N_r=303, N_r/N_b=0.8682)
Train Coverage: (N_b=6789, N_r=5846, N_r/N_b=0.8611)

[7] intergovorgs3(X1, X2) :- reintergovorgs(X1, X2) and not(economicaid(X2, X3)) and
    intergovorgs(X2, X1) and not(eemigrants(X3, X2)) and timesinceally(X2, X1).
Training Acc: 0.6106 | Eval Acc: 0.6163

```